AI-Based Image Classification System for Coconut Germplasm Identification

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Project Proposal Report

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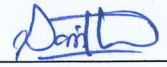
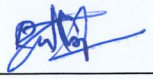
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Coconut (Cocos nucifera L.) is a major plantation crop in Sri Lanka and plays an important role in the country's agricultural economy and rural livelihoods. The Coconut Research Institute (CRI) maintains gene bank collections to preserve diverse coconut germplasm varieties used for breeding and research. However, germplasm identification is currently performed through manual observation of morphological traits such as crown structure, leaf color, palm height, and nut shape, which is time-consuming and difficult to scale across large collections and plantations. In addition, farmers and plantation owners often lack the expertise required to accurately identify coconut germplasm varieties in their fields. Although deep learning and computer vision techniques have been widely applied in agriculture for plant disease detection and plant classification, limited research has focused on automated identification of coconut germplasm varieties using image-based methods. To address this gap, this study proposes an AI-based image classification system that allows users, including farmers and agricultural researchers, to upload images of coconut palms and nuts and automatically identify germplasm types. The system uses Convolutional Neural Networks (CNNs) with transfer learning, few-shot learning for limited datasets, and Grad-CAM explainable AI for model interpretation. A secure web-based platform developed using FastAPI and PostgreSQL with role-based access control will enable image upload, automated classification, and secure storage of identification results. The proposed system aims to improve the efficiency, accessibility, and accuracy of coconut germplasm identification while supporting both agricultural research and practical plantation management.

Keywords: Coconut Germplasm Identification, Deep Learning, Computer Vision, Few-Shot Learning, Explainable AI.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

CNN – Convolutional Neural Network

CRI – Coconut Research Institute

FAO – Food and Agriculture Organization

FAOSTAT – Food and Agriculture Organization Statistical Database

Grad-CAM – Gradient-weighted Class Activation Mapping

ICCV – International Conference on Computer Vision

ICIP – International Conference on Image Processing

ML – Machine Learning

RBAC – Role-Based Access Control

SaaS – Software as a Service

SQL – Structured Query Language

1. INTRODUCTION

1.1 Background and Context

Coconut (Cocos nucifera L.) is one of the most important crops grown on plantations in Sri Lanka. It is also very important for the country's agricultural economy, food security, and rural livelihoods. Sri Lanka is known as one of the top coconut-producing countries. The crop is very important for both domestic use and exports, such as coconut oil, desiccated coconut, and coconut-based goods [1]. To boost productivity, develop varieties that withstand climate change, and ensure agriculture can grow sustainably, it's important to preserve coconut genetic diversity.

The Coconut Research Institute (CRI) of Sri Lanka is the main national organization in-charge of coconut research, development, and the protection of coconut genetic resources. To protect a wide range of coconut germplasm types, including tall dwarf and hybrid types, CRI has several regional gene banks in different agro-climatic zones across the country [2]. These germplasm collections are useful for breeding programs, scientific research, and protecting biodiversity over the long term.

Accurate identification and documentation of coconut germplasm are crucial for effective gene bank management. In CRI gene banks, germplasm identification has historically been carried out by manually observing morphological traits like leaf color, crown structure, nut shape, and nut color. Despite being widely used in agricultural research, these conventional techniques mainly rely on field-based observations and expert knowledge. Because of this, managing large germplasm collections across several regional gene banks can be a time-consuming, subjective, and challenging process [3].

New possibilities for automating agricultural analysis tasks have been made possible by recent developments in computer vision and artificial intelligence (AI). Convolutional Neural Networks (CNNs), one type of deep learning technique, have proven particularly effective in image classification applications, such as crop disease detection, plant species identification, and agricultural monitoring [4]. Furthermore, classification models can function well even with small labelled datasets thanks to methods like few-shot learning, which is a frequent problem in agricultural research [5]. Additionally, explainable AI methods like Grad-CAM can offer visual interpretations of model predictions, enhancing experts' trust and transparency in AI-based systems [6].

However, the use of AI-based image classification for identifying the type of coconut germplasm in Sri Lanka is still in its infancy. Most farmers and plantation owners cannot easily identify the type of coconut germplasm in their plantations. The available technology can only be used in institutional environments for research purposes. Hence, this research aims to develop an AI-based image classification system that can help users identify the type of coconut germplasm by uploading images of the coconut plant or nut. The main goal of this system is to create a digital platform that can be useful for both agricultural research and practical purposes.

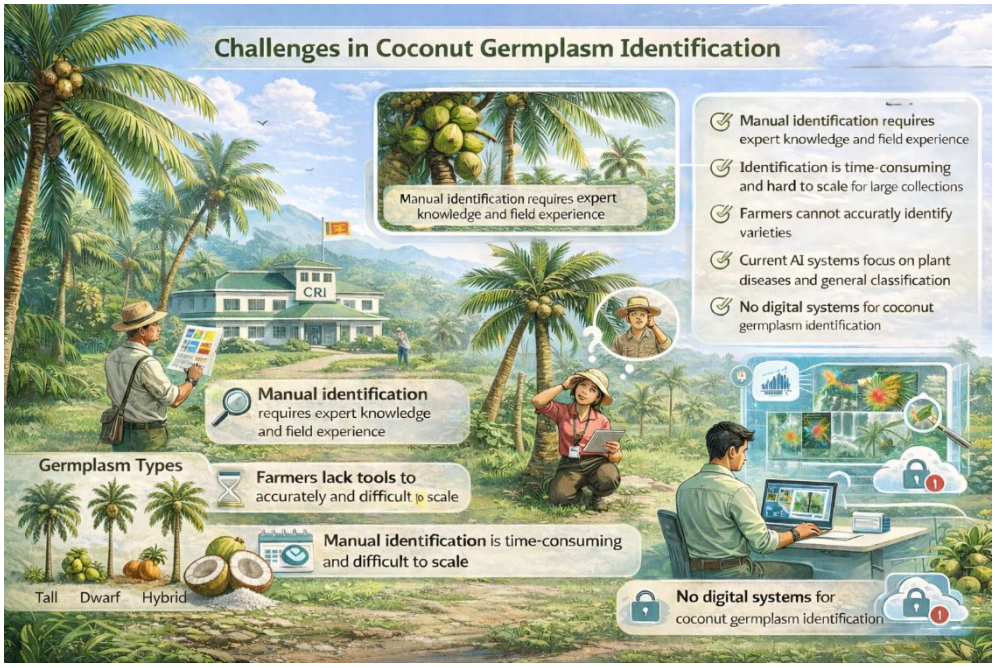
1.2 Research Problem

Coconut germplasm identification is an important process for maintaining coconut genetic resources and supporting breeding programs. However, several challenges exist in the current identification process.

- Coconut germplasm identification is mainly performed through manual observation of morphological characteristics, which requires expert knowledge and field experience.
- The manual identification process is time-consuming and difficult to scale, especially when large numbers of coconut palms must be evaluated across different plantations or gene bank collections.
- Farmers and plantation owners often cannot accurately identify coconut varieties in their plantations because they do not have access to expert knowledge or research facilities.
- Most existing agricultural AI systems focus on plant disease detection or general plant classification, rather than identifying specific coconut germplasm varieties.
- There is a lack of accessible digital systems that allow farmers or researchers to upload coconut images and automatically identify germplasm types using artificial intelligence.

Therefore, an intelligent image-based system is needed to automatically classify coconut germplasm varieties using deep learning techniques. Such a system would support both agricultural researchers and farmers by providing faster, more accessible, and data-driven germplasm identification.

Figure 1: Challenges in Coconut Germplasm Identification



1.3 Literature Review

Automated image analysis in agriculture has been greatly enhanced by recent developments in computer vision and artificial intelligence. In large-scale image classification tasks and visual pattern recognition, Convolutional Neural Networks (CNNs) have shown excellent performance. Deep learning is a potent method for image classification problems, as demonstrated by Krizhevsky et al.'s demonstration that deep CNN architectures can efficiently learn hierarchical visual features from images [4].

Deep learning methods have been used in a number of studies to identify plants automatically. Deep learning models can successfully classify plant species based on visual characteristics, as shown by Lee et al.'s CNN-based framework for plant species recognition using leaf images [9]. Similar to this, Carranza-Rojas et al. examined automated plant identification using photos of herbarium specimens and showed how deep learning could be used for large-scale plant recognition and biodiversity monitoring [10].

Some research has looked into automated crop variety identification, which is closely related to germplasm classification, in addition to general plant identification. Using grain image datasets, Chen et al. created a deep learning method for rice grain variety identification using convolutional neural networks and attained high classification accuracy [11]. In a similar vein, Pourreza et al. used machine learning and image processing methods to identify wheat seed varieties using morphological characteristics taken from seed photos [12]. These studies show that distinct crop varieties can be identified using visual morphological traits.

However, training deep learning models is difficult because agricultural datasets are frequently small. By allowing models to learn new classes with a limited number of training examples, few-shot learning techniques have been proposed to address this problem [5]. In agricultural applications, where gathering large labelled datasets is challenging, these methods are especially helpful.

Model interpretability is a crucial component of AI-based classification systems. Because deep learning models are frequently regarded as "black-box" systems, it can be challenging for subject matter experts to comprehend how predictions are made. Grad-CAM, an explainable AI method that produces visual heatmaps emphasizing the image regions that affect CNN predictions, was developed by Selvaraju et al. to address this problem [6]. Expert interpretation and validation of model predictions is made possible by this technique, which also increases transparency.

Although previous research has demonstrated the potential of deep learning for plant identification and crop variety classification, very limited work has focused on automated coconut germplasm identification using image-based methods. Furthermore, existing studies rarely integrate explainable AI techniques and secure digital platforms for managing germplasm records. Therefore, there is a need for an intelligent system that integrates deep learning, few-shot learning, explainable AI, and secure digital data management to support automated coconut germplasm identification.

Study	Method	Application	Limitation
Krizhevsky et al. [4]	Convolutional Neural Networks (CNN)	Large-scale image classification *(plant type: general image dataset – ImageNet)*	Not applied to agricultural crop identification
Lee et al. [9]	Deep CNN	Automated plant species identification using leaf images *(plant type: multiple plant species)*	Does not focus on crop variety or germplasm identification
Carranza-Rojas et al. [10]	Deep learning	Large-scale plant identification for biodiversity monitoring *(plant type: herbarium plant specimens)*	Not designed for crop variety classification
Chen et al. [11]	Deep CNN	Automatic rice variety identification using grain images *(plant type: rice)*	Limited to a specific crop dataset
Pourreza et al. [12]	Image processing + Machine learning	Wheat seed variety identification using morphological features *(plant type: wheat)*	Traditional machine learning methods with limited scalability
Proposed System	CNN + Few-shot learning + Grad-CAM	Automated coconut germplasm identification and digital gene-bank record generation *(plant type: coconut palms and nuts)*	—

Table 1: Literature Review

1.4 Research Gap

Recent studies have demonstrated the effectiveness of deep learning and computer vision techniques for automated plant image analysis. Convolutional Neural Networks (CNNs) have been widely used for image classification tasks and have shown strong performance in visual feature extraction and recognition [4]. Several researchers have applied these techniques to plant identification problems. For example, Lee et al. developed a deep learning-based approach for plant species identification using leaf images [9], while Carranza-Rojas et al. demonstrated automated plant recognition using herbarium specimen images for biodiversity monitoring [10].

In addition to plant species recognition, some studies have explored crop variety identification using image-based methods. For instance, Chen et al. proposed a deep learning model for identifying rice grain varieties using convolutional neural networks [11]. Similarly, Pourreza et al. applied image processing techniques to identify wheat seed varieties based on morphological characteristics extracted from seed images [12]. These studies demonstrate that visual morphological features can be used to distinguish between different crop varieties.

However, most existing research focuses on general plant species identification or crop variety classification for specific crops such as rice and wheat, rather than identifying coconut germplasm varieties. At present, coconut germplasm identification at the Coconut Research Institute (CRI) of Sri Lanka is mainly performed through manual evaluation of morphological characteristics such as crown structure, nut shape, and leaf appearance by field experts [2]. This manual approach requires extensive domain knowledge, is time-consuming, and becomes difficult to scale across large gene bank collections.

Furthermore, existing plant classification systems typically focus only on prediction accuracy and do not incorporate explainable AI techniques that allow researchers to interpret model predictions. In addition, many current systems do not provide secure digital platforms for storing germplasm records with controlled access and traceability, which are essential for institutional research databases.

Although previous studies demonstrate the effectiveness of deep learning techniques for plant identification and crop classification, very limited research focuses specifically on automated coconut germplasm identification using field images. Additionally, most existing agricultural AI systems are designed for research environments and are not accessible to farmers or plantation owners who need simple tools for identifying coconut varieties in their own fields. In Sri Lanka, germplasm identification still relies heavily on expert-based manual observation, which limits accessibility and scalability. Furthermore, existing systems rarely integrate explainable AI techniques and user-friendly digital platforms that allow non-experts to understand model predictions. Therefore, there is a need for an intelligent and accessible system that enables farmers and researchers to upload coconut images, automatically identify germplasm types using deep learning techniques, and provide interpretable predictions through explainable AI methods.

Study / System	Image Classification	Germplasm Identification	Few-Shot Learning	Explainable AI	Secure Database
Krizhevsky et al. (2012) – CNN Image Classification [4]	✓ Deep CNN image classification	✗ Not focused on plant varieties	✗ Not used	✗ No explainability	✗ No database
Lee et al. (2015) – Deep Plant Identification [9]	✓ Plant species identification	✗ Not focused on crop germplasm	✗ Not used	✗ No explainable AI	✗ No database
Carranza-Rojas et al. (2017) – Herbarium Identification [10]	✓ Automated plant identification	✗ Not focused on crop varieties	✗ Not used	✗ No model interpretation	✗ No secure database
Chen et al. (2019) – Rice Variety Identification [11]	✓ CNN-based grain image classification	✓ Rice variety identification	✗ Not used	✗ No explainable AI	✗ No database
Pourreza et al. (2012) – Wheat Seed Identification [12]	✓ Seed image classification	✓ Wheat variety identification	✗ Not used	✗ No explainability	✗ No database
Proposed System	✓ Coconut palm & nut image classification	✓ Coconut germplasm identification	✓ Few-shot learning	✓ Grad-CAM explainability	✓ Secure digital gene-bank database

Table 2: Comparison of Existing Systems and Proposed System

2. OBJECTIVE

2.1 Main Objective

To develop a secure AI-based image classification system for automated coconut germplasm identification using coconut palm and nut images, capable of generating structured digital gene bank records to support the Coconut Research Institute of Sri Lanka.

2.2 Specific Objectives

1. Collect and prepare an image dataset of coconut palms and nuts from research institutes and farmer plantations for training and evaluating the deep learning model.
2. Develop a Convolutional Neural Network (CNN)-based image classification model to identify coconut germplasm varieties using visual features extracted from coconut palm and nut images.
3. Apply transfer learning and few-shot learning techniques to improve classification performance when labeled datasets are limited.
4. Implement explainable artificial intelligence techniques such as Grad-CAM to visualize and interpret the image regions influencing the classification results.
5. Design and develop a user-friendly web-based platform that allows farmers and agricultural researchers to upload coconut images and obtain automated germplasm identification results.
6. Store and manage classification results, image records, and related metadata in a centralized database to support research analysis and future dataset expansion.

3. METHODOLOGY

This study uses a Design Science Research methodology, which focuses on creating a technological solution to a practical issue. Using images of coconut palms and nuts gathered from gene bank locations maintained by the Coconut Research Institute (CRI) of Sri Lanka, the project seeks to create a secure AI-based image classification system that can identify coconut germplasm varieties.

During the data collection phase, images of coconut palms and nuts will be collected from multiple sources, including research institutions, agricultural fields, and farmer plantations. These images will represent different coconut germplasm varieties and environmental conditions to ensure dataset diversity. Each image will be associated with metadata such as region, possible variety, and plantation details when available. Ethical considerations will be addressed by ensuring that collected images are used only for research purposes and stored securely. Computer vision methods will be used to standardize and improve the dataset during the image preprocessing phase. To enhance model training performance, image processing tasks like resizing, normalization, and noise reduction will be carried out using the OpenCV library.

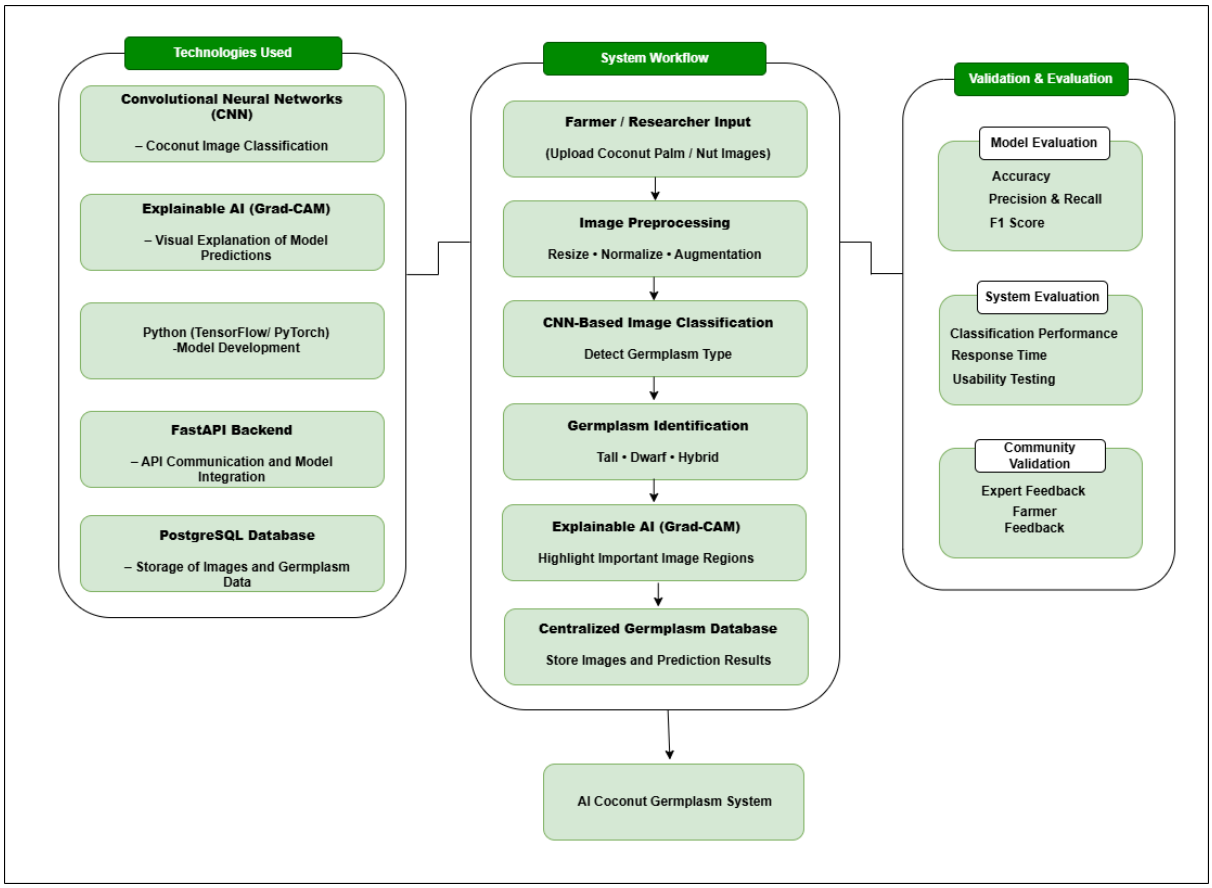
A Convolutional Neural Network (CNN) architecture will be used to automatically extract visual features from images of coconut palms and nuts for classification. Transfer learning will be combined with pretrained deep learning models, like EfficientNet or MobileNet, to increase classification accuracy while cutting training time. Few-shot learning techniques will be used to allow the model to classify germplasm varieties using a small number of training examples, as agricultural datasets are frequently limited.

Grad-CAM explainable AI techniques will be used to create visual heatmaps that highlight the image regions that affect classification decisions in order to improve model interpretability. This will enable CRI specialists to verify and analyze the model's forecasts.

The system will be put into use as a web-based platform that enables authorized users to upload pictures of coconut palms and nuts and receive automated classification results. Python-based frameworks like FastAPI will be used to develop the backend, and a centralized PostgreSQL database will hold the metadata and classification results. To guarantee safe access and traceability of system operations, role-based access control systems will be put into place.

Lastly, common machine learning metrics like accuracy, precision, recall, and F1-score will be used to assess the performance of the suggested model. Experimental validation will be conducted using a separate test dataset to assess the effectiveness of the proposed system in identifying coconut germplasm varieties.

Figure 2:System Methodology Flow



4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is an AI-based image classification platform designed for automated identification of coconut germplasm varieties using images of coconut palms and nuts. The system allows users such as farmers, agricultural officers, and researchers to upload coconut images through a web or mobile interface. The uploaded images are processed by the system to identify the germplasm type based on visual morphological features extracted from the images.

The system starts with the Image Input Module, where authorized CRI staff or farmer use a web interface to upload pictures of coconut palm crowns and nuts. Every image has a distinct Palm ID, region, and associated metadata. The Image Preprocessing Module then processes the uploaded images, performing quality validation, resizing, and normalization to guarantee consistent input for the model.

The CNN-based Germplasm Classification Module receives the processed images and uses visual patterns learned from palm and nut images to identify the types of coconut germplasm. When there are few labelled samples available, a Few-Shot Learning module is incorporated to enhance classification. Concurrently, CRI-aligned visual characteristics like leaf color, nut color, crown density, and nut shape are obtained through a Morphological Parameter Extraction step. To display the image regions that affected the prediction, an Explainable AI module based on Grad-CAM creates visual heatmaps.

The predicted germplasm type, extracted visual parameters, confidence score, and Grad-CAM explanation are the final products of the interaction between these modules' outputs. The Secure Record Generation Module receives these outputs, transforms them into organized digital gene-bank records, and stores them in a centralize database.

Role-Based Access Control (RBAC) and audit logging are used by the system to manage permissions and track system actions like uploads, predictions, and verification in order to guarantee secure operation. While preserving centralized and secure data management, the modular design facilitates scalability across several CRI gene-bank locations.

5.USER REQUIREMENTS

Stakeholder Identification

The proposed system will involve several stakeholders who interact with the coconut germplasm identification process.

Farmers / Plantation Owners – upload coconut palm or nut images from plantations to identify germplasm types.

Agricultural Researchers – analyze classification results and improve dataset quality.

Agricultural Officers – assist farmers in identifying coconut varieties and verifying results.

System Administrators – manage user accounts, system security, and database management.

Research Team – develop, monitor, and maintain the AI-based system.

Functional Requirements

1. The system shall allow authorized users to upload coconut palm and nut images.
2. The system shall assign each uploaded record to a unique Palm ID.
3. The system shall preprocess uploaded images before classification.
4. The system shall classify coconut germplasm types using CNN-based models.
5. The system shall support few-shot learning for low-data germplasm classes.
6. The system shall extract CRI-aligned visual morphological parameters from the images.
7. The system shall generate Grad-CAM visual explanations for model predictions.
8. The system shall store predictions, extracted parameters, and metadata in a centralized database.
9. The system shall allow CRI experts to verify or update prediction results.
10. The system shall maintain logs of uploads, predictions, and verification activities.

Non-Functional Requirements

Security Requirements

- The system shall implement role-based access control for different user roles.
- The system shall protect stored records and user activities through secure authentication and audit logging.

Performance Requirements

- The system shall process uploaded images and generate predictions within an acceptable response time.

- The system shall support efficient storage and retrieval of digital gene-bank records.

Usability Requirements

- The system shall provide a simple and user-friendly interface for CRI personnel.
- The system shall allow easy image upload, result viewing, and record verification.

Reliability Requirements

- The system shall maintain consistent performance during repeated image classification tasks.
- The system shall preserve digital records accurately without data loss.

Scalability Requirements

- The system shall support expansion to multiple CRI regional gene-bank locations.
- The system shall allow future extension to additional germplasm classes and image datasets.

6. COMMERCIALIZATION PLAN

The proposed AI-based coconut germplasm identification system can be used by multiple stakeholders including farmers, agricultural researchers, plantation companies, and agricultural institutions such as the Coconut Research Institute (CRI). Farmers can use the system to identify coconut varieties in their plantations by simply uploading coconut palm or nut images, while researchers can use the collected data for germplasm analysis and research purposes.

By introducing an automated image-based germplasm identification platform that employs deep learning techniques to analyze images of coconut palms and nuts and produce structured digital gene-bank records, the system offers a compelling value proposition. Compared to conventional manual identification techniques, this method increases consistency, accuracy, and efficiency.

Several CRI regional gene-bank locations can access the platform as a centralized web-based system. In terms of commercialization, the system might be made available to plantation companies and agricultural research institutions through subscription-based access (SaaS) or institutional licensing.

The creation of AI models, gathering datasets, cloud infrastructure, and system upkeep are the primary cost components. A pricing strategy might include yearly institutional subscriptions or licensing agreements based on these expenses.

The suggested system has a number of competitive advantages, such as explainable AI using Grad-CAM to increase transparency, support for small datasets through few-shot learning, and automated germplasm identification using CNN-based models. Software copyrights and institutional intellectual property policies can also safeguard software components and AI models, guaranteeing correct ownership and data governance.

7. BUDGET AND JUSTIFICATION

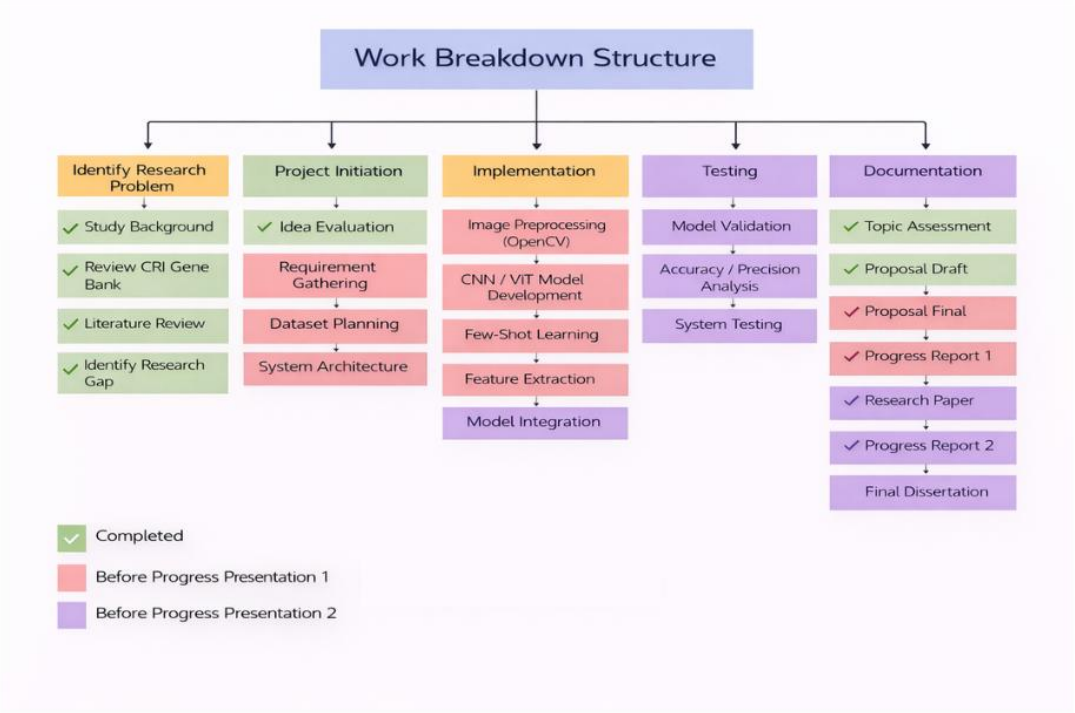
Resources	**Estimated Price (LKR)**
Cloud server hosting (AI model deployment and database storage)	20,000.00
Transportation (field visits to CRI gene banks for image data collection)	12,000.00
Internet and data usage (dataset transfer, cloud access, development work)	5,000.00
Stationery and documentation	2,000.00
Hardware components/storage devices (external drives, minor hardware upgrades)	13,000.00
Miscellaneous expenses	5,000.00
Total	**57,000.00**

Table 3:Expected Expenditure

The project is expected to cost LKR 57,000. Cloud server hosting, which is necessary to store image datasets, implement the AI classification model, and manage the centralized database for germplasm records, accounts for the majority of the expense. For field visits to the Coconut Research Institute (CRI) gene bank locations to gather images of coconut palms and nuts for dataset preparation, transportation costs are necessary. Stationery for documentation, hardware storage devices for managing large image datasets, and internet and data usage for cloud access and dataset transfer are additional costs. The development, testing, and implementation of the suggested AI-based coconut germplasm identification system depend on these resources.

8. WORK BREAKDOWN STRUCTURE (WBS)

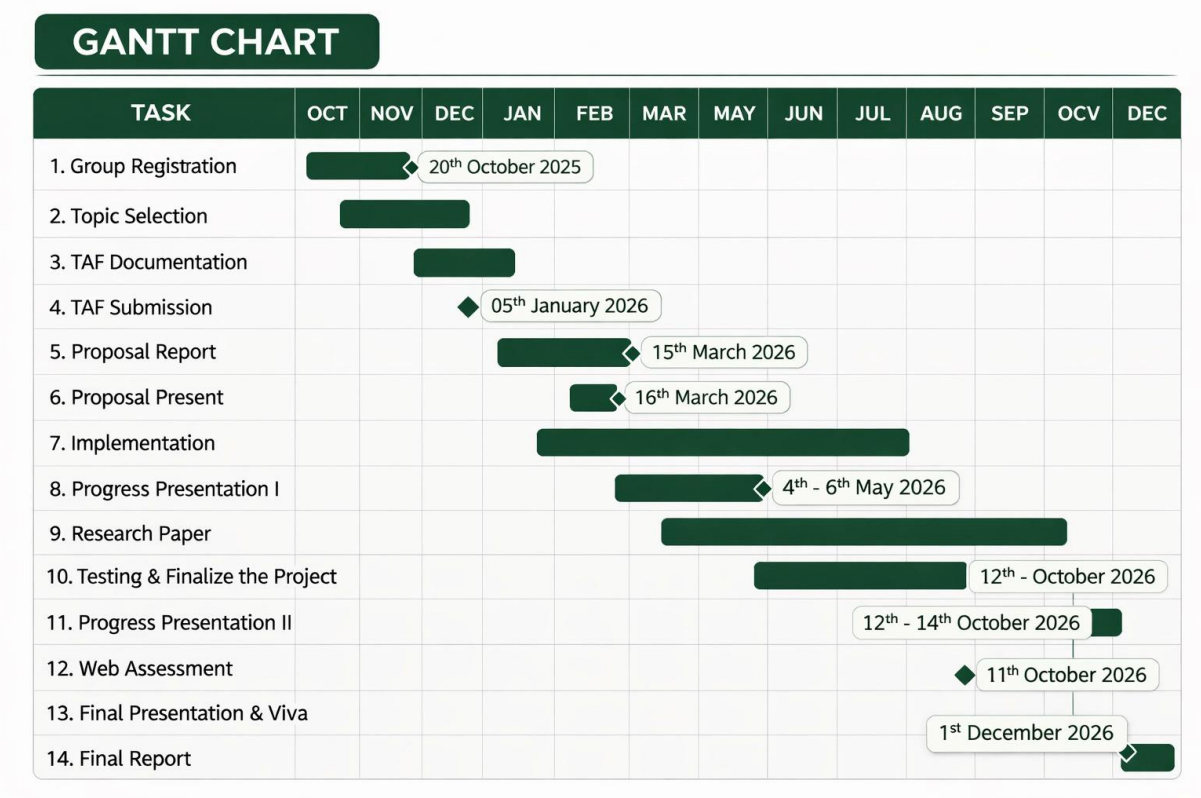
Figure 3: Work-Breakdown Structure



The work breakdown structure for the research project has five main stages: identifying the research problem, project initiation, implementation, testing, and documentation. During the first stage, the analysis of the research problem will be done, considering the background of the identification of the coconut germplasm, the gene bank practice, and the literature review, as well as the identification of the gaps in the existing identification process. During the project initiation stage, the analysis of the project idea will be done, the requirements of the system will be gathered, the process of collecting the data set will be planned, and the overall system architecture will be designed. During the implementation stage, the image preprocessing techniques will be applied to the coconut palm and the nut image, followed by the development of the deep learning models for the classification and feature extraction of the images. During the testing stage, the performance of the model will be done, considering the analysis of the model's performance, the precision analysis, and the testing of the system's workflow. Finally, the documentation stage will be done, considering the proposal documents, the progress reports, the research paper, and the dissertation.

9. GANTT CHART

Figure 4: Gantt Chart



10. REFERENCES LIST

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